

CS 2601 Linear and Convex Optimization

6. Gradient descent (part 3)

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Step size

Gradient descent

$$\mathbf{x}_{k+1} = \mathbf{x}_k - t_k \nabla f(\mathbf{x}_k)$$

- constant step size: $t_k = t$ for all k
- exact line search: optimal t_k for each step

$$t_k = \arg \min_s f(\mathbf{x}_k - s \nabla f(\mathbf{x}_k))$$

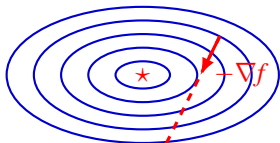
- backtracking line search (Armijo's rule): t_k satisfies

$$f(\mathbf{x}_k) - f(\mathbf{x}_k - t_k \nabla f(\mathbf{x}_k)) \geq \alpha t_k \|\nabla f(\mathbf{x}_k)\|_2^2$$

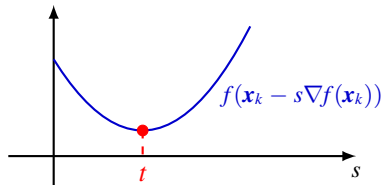
for some given $\alpha \in (0, 1)$.

Exact line search

- 1: initialization $\mathbf{x} \leftarrow \mathbf{x}_0 \in \mathbb{R}^n$
- 2: **while** $\|\nabla f(\mathbf{x})\| > \delta$ **do**
- 3: $t \leftarrow \arg \min_s f(\mathbf{x} - s \nabla f(\mathbf{x}))$
- 4: $\mathbf{x} \leftarrow \mathbf{x} - t \nabla f(\mathbf{x})$
- 5: **end while**
- 6: **return** $\mathbf{x}$



level curves of $f(x_1, x_2) = \frac{x_1^2}{4} + x_2^2$



Note. Often impractical; used only if the inner minimization is cheap.

Exact line search for quadratic functions

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{b}^T \mathbf{x}, \quad \mathbf{Q} \succ \mathbf{O}$$

- gradient at $\mathbf{x}_k$ is $\mathbf{g}_k = \nabla f(\mathbf{x}_k) = \mathbf{Q} \mathbf{x}_k + \mathbf{b}$
- second-order Taylor expansion is exact for quadratic functions,

$$\begin{aligned} h(t) &= f(\mathbf{x}_k - t \mathbf{g}_k) \\ &= f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T (-t \mathbf{g}_k) + \frac{1}{2} (-t \mathbf{g}_k)^T \nabla^2 f(\mathbf{x}_k) (-t \mathbf{g}_k) \\ &= \left(\frac{1}{2} \mathbf{g}_k^T \mathbf{Q} \mathbf{g}_k \right) t^2 - \mathbf{g}_k^T \mathbf{g}_k t + f(\mathbf{x}_k) \end{aligned}$$

- minimizing $h(t)$ yields best step size

$$t_k = \frac{\mathbf{g}_k^T \mathbf{g}_k}{\mathbf{g}_k^T \mathbf{Q} \mathbf{g}_k}$$

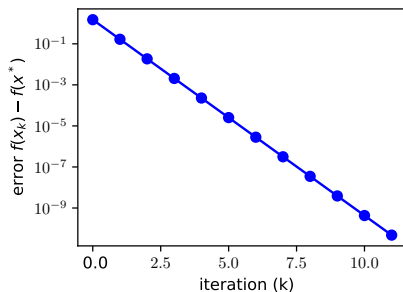
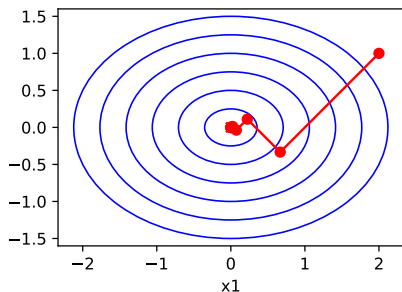
- update step

$$\mathbf{x}_{k+1} = \mathbf{x}_k - t_k \mathbf{g}_k = \mathbf{x}_k - \frac{\mathbf{g}_k^T \mathbf{g}_k}{\mathbf{g}_k^T \mathbf{Q} \mathbf{g}_k} \mathbf{g}_k$$

Example

$$f(x_1, x_2) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} = \frac{\gamma}{2} x_1^2 + \frac{1}{2} x_2^2, \quad \mathbf{Q} = \text{diag}\{\gamma, 1\}$$

Well-conditioned. $\gamma = 0.5$, $\mathbf{x}_0 = (2, 1)$



Fast convergence.

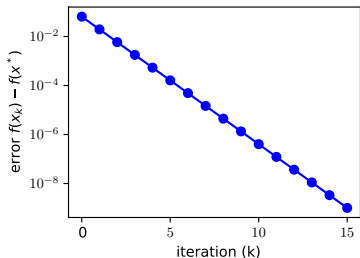
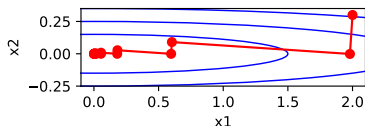
Note. Successive gradient directions are always orthogonal, as

$$0 = h'(t_k) = -\nabla f(\mathbf{x}_k - t_k \nabla f(\mathbf{x}_k))^T \nabla f(\mathbf{x}_k) = -\nabla f(\mathbf{x}_{k+1})^T \nabla f(\mathbf{x}_k)$$

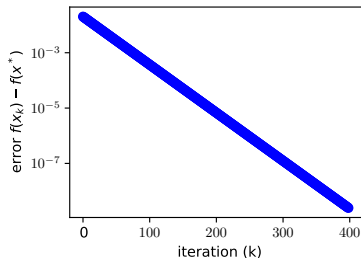
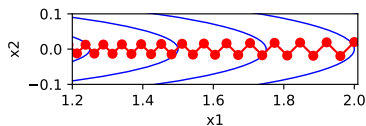
Example (cont'd)

$$f(x_1, x_2) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} = \frac{\gamma}{2} x_1^2 + \frac{1}{2} x_2^2, \quad \mathbf{Q} = \text{diag}\{\gamma, 1\}$$

Ill-conditioned. $\gamma = 0.01$, convergence rate depends on initial point



$\mathbf{x}_0 = (2, 0.3)$, fast convergence



$\mathbf{x}_0 = (2, 0.02)$, slow convergence

Convergence analysis

Theorem. If f is m -strongly convex and L -smooth, and $\mathbf{x}^*$ is a minimum of f , then the sequence $\{\mathbf{x}_k\}$ produced by gradient descent with exact line search satisfies

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \left(1 - \frac{m}{L}\right)^k [f(\mathbf{x}_0) - f(\mathbf{x}^*)]$$

Notes.

- $0 \leq 1 - \frac{m}{L} < 1$, so $f(\mathbf{x}_k) \rightarrow f(\mathbf{x}^*)$ exponentially fast
- $\frac{m}{2} \|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq f(\mathbf{x}_k) - f(\mathbf{x}^*)$ by strong convexity, so $\mathbf{x}_k \rightarrow \mathbf{x}^*$ exponentially fast
- The number of iterations to reach $f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \epsilon$ is $O(\log \frac{1}{\epsilon})$. For $\epsilon = 10^{-p}$, $k = O(p)$, linear in the number of significant digits.
- The convergence rate depends on the condition number L/m and can be slow if L/m is large. When close to $\mathbf{x}^*$, we can estimate L/m by $\kappa(\nabla^2 f(\mathbf{x}^*))$.

Proof

Similar to slide 12 of §6 part 2, with a modified first step (highlighted).

1. Lower bound the improvement in the k -th iteration

- By the quadratic upper bound for L -smooth functions,

$$f(\mathbf{x}_k - t\nabla f(\mathbf{x}_k)) - f(\mathbf{x}_k) \leq -t\left(1 - \frac{Lt}{2}\right)\|\nabla f(\mathbf{x}_k)\|^2$$

- Minimize over t on both sides,

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}_k) \leq -\frac{1}{2L}\|\nabla f(\mathbf{x}_k)\|^2 \quad (\dagger)$$

2. Upper bound the suboptimality gap (slide 8 of §6 part 2),

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \frac{1}{2m}\|\nabla f(\mathbf{x}_k)\|^2 \quad (\ddagger)$$

3. Eliminate $\|\nabla f(\mathbf{x}_k)\|$ from $(\dagger)$ and $(\ddagger)$,

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq \left(1 - \frac{m}{L}\right) [f(\mathbf{x}_k) - f(\mathbf{x}^*)]$$

Backtracking line search

Exact line search is often expensive and not worth it. Suffices to find a good enough step size. One way to do so is to use **backtracking line search** with **Armijo's rule**.

Gradient descent with backtracking line search

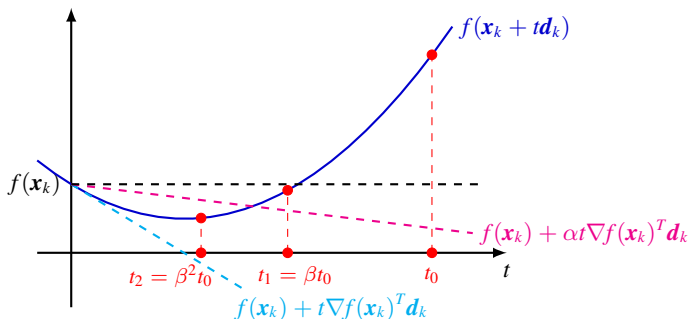
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1: initialization  $\mathbf{x} \leftarrow \mathbf{x}_0 \in \mathbb{R}^n$ 
2: while  $\|\nabla f(\mathbf{x})\| > \delta$  do
3:    $t \leftarrow t_0$ 
4:   while  $f(\mathbf{x} - t\nabla f(\mathbf{x})) > f(\mathbf{x}) - \alpha t \|\nabla f(\mathbf{x})\|_2^2$  do
5:      $t \leftarrow \beta t$ 
6:   end while
7:    $\mathbf{x} \leftarrow \mathbf{x} - t\nabla f(\mathbf{x})$ 
8: end while
9: return  $\mathbf{x}$ 
```

$\alpha \in (0, 1)$ and $\beta \in (0, 1)$ are constants. Armijo used $\alpha = \beta = 0.5$

Values suggested in [BV]: $\alpha \in [0.01, 0.3]$, $\beta \in [0.1, 0.8]$

Note. For general $\mathbf{d}$, use condition $f(\mathbf{x} + t\mathbf{d}) > f(\mathbf{x}) + \alpha t \nabla f(\mathbf{x})^T \mathbf{d}$

Backtracking line search (cont'd)



- $\nabla f(\mathbf{x}_k)^T \mathbf{d}_k < 0$ for descent direction $\mathbf{d}_k$
- start from some “large” step size t_0 ([BV] uses $t_0 = 1$)
- reduce step size geometrically until decrease is “large enough”

$$f(\mathbf{x}_k) - f(\mathbf{x}_k + t \mathbf{d}_k) \geq \alpha t (-\nabla f(\mathbf{x}_k)^T \mathbf{d}_k)$$

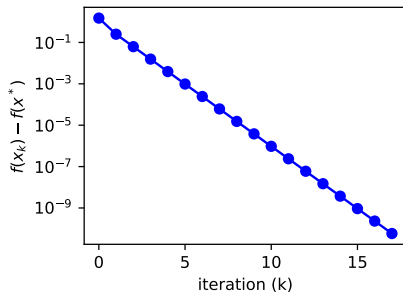
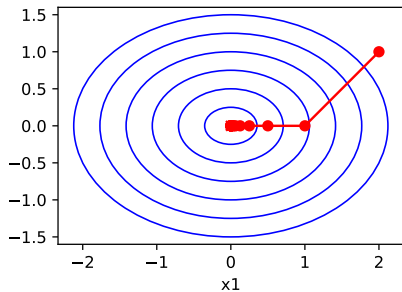
cf. descent lemma for $\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$ and constant step size

$$f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) = f(\mathbf{x}_k) - f(\mathbf{x}_k - t \nabla f(\mathbf{x}_k)) \geq \frac{1}{2} t \|\nabla f(\mathbf{x}_k)\|^2$$

Example

$$f(x_1, x_2) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} = \frac{\gamma}{2} x_1^2 + \frac{1}{2} x_2^2, \quad \mathbf{Q} = \text{diag}\{\gamma, 1\}$$

Well-conditioned. $\gamma = 0.5$, $\mathbf{x}_0 = (2, 1)$

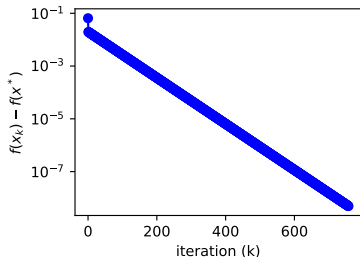
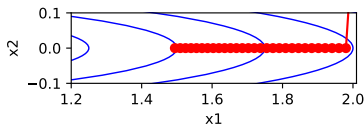


Fast convergence.

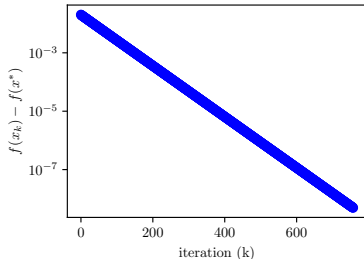
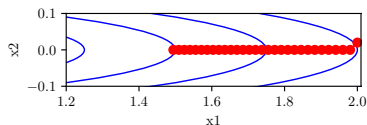
Example (cont'd)

$$f(x_1, x_2) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} = \frac{\gamma}{2} x_1^2 + \frac{1}{2} x_2^2, \quad \mathbf{Q} = \text{diag}\{\gamma, 1\}$$

Ill-conditioned. $\gamma = 0.01$



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Convergence analysis

Theorem. If f is m -strongly convex and L -smooth, and $\mathbf{x}^*$ is a minimum of f , then the sequence $\{\mathbf{x}_k\}$ produced by gradient descent with backtracking line search satisfies

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq c^k [f(\mathbf{x}_0) - f(\mathbf{x}^*)]$$

where

$$c = 1 - \min \left\{ 2m\alpha t_0, \frac{4m\beta\alpha(1-\alpha)}{L} \right\}$$

Notes.

- $c \in (0, 1)$, as

$$\frac{4m\beta\alpha(1-\alpha)}{L} \leq \frac{\beta m}{L} \leq \beta < 1$$

so $\mathbf{x}_k \rightarrow \mathbf{x}^*$ and $f(\mathbf{x}_k) \rightarrow f(\mathbf{x}^*)$ exponentially fast

- Number of iterations to reach $f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \epsilon$ is $O(\log \frac{1}{\epsilon})$. For $\epsilon = 10^{-p}$, $k = O(p)$, linear in the number of significant digits.

Proof

The inner loop terminates with a step size bounded from below.

1. By the quadratic upper bound for L -smooth functions,

$$f(\mathbf{x}_k - t\nabla f(\mathbf{x}_k)) \leq f(\mathbf{x}_k) - t(1 - \frac{Lt}{2})\|\nabla f(\mathbf{x}_k)\|^2$$

2. The inner loop terminates for sure if

$$-t(1 - \frac{Lt}{2})\|\nabla f(\mathbf{x}_k)\|^2 \leq -\alpha t\|\nabla f(\mathbf{x}_k)\|^2 \implies t \leq \frac{2(1 - \alpha)}{L}$$

3. The step size in backtracking line search satisfies

$$t_k \geq \eta \triangleq \min \left\{ t_0, \frac{2\beta(1 - \alpha)}{L} \right\}$$

- ▶ $t_k = t_0$ if Armijo's condition is satisfied by t_0
- ▶ otherwise, $\frac{t_k}{\beta} > \frac{2(1 - \alpha)}{L}$, since the inner loop did not terminate at $\frac{t_k}{\beta}$

Proof (cont'd)

Now we look at the outer loop

4. Lower bound the improvement in the k -th iteration

- By Armijo's condition in the inner loop,

$$f(\mathbf{x}_{k+1}) = f(\mathbf{x}_k - t_k \nabla f(\mathbf{x}_k)) \leq f(\mathbf{x}_k) - \alpha t_k \|\nabla f(\mathbf{x}_k)\|^2$$

- Since $t_k \geq \eta$ by step 3,

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq f(\mathbf{x}_k) - f(\mathbf{x}^*) - \alpha \eta \|\nabla f(\mathbf{x}_k)\|^2$$

5. Upper bound the suboptimality gap by (‡) of slide 7,

$$\|\nabla f(\mathbf{x}_k)\|^2 \geq 2m[f(\mathbf{x}_k) - f(\mathbf{x}^*)]$$

6. Eliminate $\|\nabla f(\mathbf{x}_k)\|$ from steps 4 and 5,

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq (1 - 2m\alpha\eta)[f(\mathbf{x}_k) - f(\mathbf{x}^*)] = c[f(\mathbf{x}_k) - f(\mathbf{x}^*)]$$

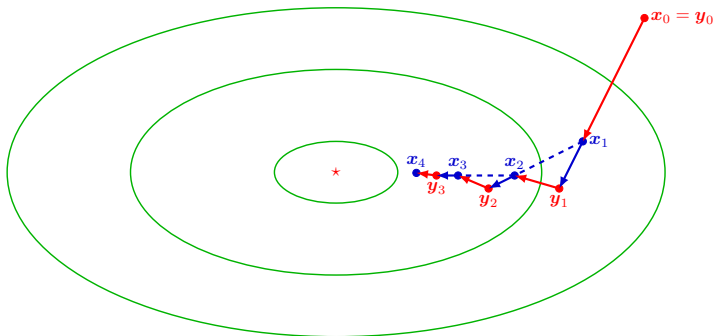
so

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq c^k[f(\mathbf{x}_0) - f(\mathbf{x}^*)]$$

Nesterov's accelerated gradient descent (AGD)

Suppose f is L -smooth and m -strongly convex ($m \geq 0$)

- 1: initialize $\mathbf{x}_0 = \mathbf{y}_0$
- 2: **for** $k = 0, 1, 2, \dots$ **do**
- 3: $\mathbf{x}_{k+1} = \mathbf{y}_k - \frac{1}{L} \nabla f(\mathbf{y}_k)$
- 4: $\mathbf{y}_{k+1} = \mathbf{x}_{k+1} + \beta_k (\mathbf{x}_{k+1} - \mathbf{x}_k)$
- 5: **end for**



Accelerated gradient descent (cont'd)

Theorem. Suppose f is L -smooth and m -strongly convex. Let $q = m/L$, $\alpha_0 \in [\sqrt{q}, 1)$, $\alpha_{k+1} = \frac{\sqrt{(\alpha_k^2 - q)^2 + 4\alpha_k^2} + q - \alpha_k^2}{2}$ for $k \geq 0$. If $\beta_k = \frac{\alpha_k(1 - \alpha_k)}{\alpha_k^2 + \alpha_{k+1}}$ for $k \geq 0$, then

$$f(\mathbf{x}_k) - f^* \leq \min \left\{ (1 - \sqrt{q})^k, \frac{4}{(2 + k\sqrt{\gamma_0})^2} \right\} \left(f(\mathbf{x}_0) - f^* + \frac{\gamma_0}{2} \|\mathbf{x}_0 - \mathbf{x}^*\|^2 \right),$$

where $\gamma_0 = \alpha_0(\alpha_0 - q)/(1 - \alpha_0)$.

Note. The convergence rate of AGD is $O(1/k^2)$ if $m = 0$ and $O((1 - \sqrt{q})^k)$ if $m > 0$. Recall the rate of GD is $O(1/k)$ if $m = 0$ and $O((1 - q)^k)$ if $m > 0$.

Note. Nesterov also proved lower bounds for first-order methods, i.e. there exists an L -smooth f_1 , and an L -smooth and m -strongly convex f_2 s.t.

$$f_1(\mathbf{x}_k) - f^* = \Omega\left(\frac{1}{k^2}\right), \quad f_2(\mathbf{x}_k) - f^* = \Omega\left(\left(\frac{1 - \sqrt{q}}{1 + \sqrt{q}}\right)^{2k}\right).$$

Nonconvex functions

GD can also be applied to nonconvex functions, but with **no** guarantee for optimality. It only finds an approximately stationary point.

Theorem. If f is L -smooth, then for step size $t \in (0, \frac{1}{L}]$, the sequence $\{\mathbf{x}_k\}$ produced by GD satisfies

$$\min_{0 \leq i \leq k} \|\nabla f(\mathbf{x}_i)\| \leq \sqrt{\frac{2(f(\mathbf{x}_0) - f^*)}{t(k+1)}}$$

Proof. By slide 15 of §6 part 1,

$$\min_{0 \leq i \leq k} \|\nabla f(\mathbf{x}_i)\|^2 \leq \|\nabla f(\mathbf{x}_i)\|^2 \leq \frac{2}{t}(f(\mathbf{x}_i) - f(\mathbf{x}_{i+1}))$$

Summing over i from 0 to k completes the proof.

Note. The convergence rate is $O(1/\sqrt{k})$, which turns out to be optimal for deterministic algorithms for finding stationary points of functions with Lipschitz gradients.